



Quiz No. 1

CS3004

Software Design and Analysis

Total marks: 30

Total time: 40 minutes

Date: 10/02/2026

Reg. No. _____

Name. _____

Scenario: Online Library Management System

A university library allows registered members (students and faculty) to borrow items, including books, journals, and DVDs. Each member has a unique member ID, name, email, and phone number. The library maintains a catalog of items; each item has a title, author(s), publication year, ISBN/unique ID, and status (available/borrowed/reserved).

A loan is created when a member borrows one or more items. Each loan records the checkout date, due date (usually 14 days later), and any fine if returned late (calculated per day per item). A member can have multiple active loans but is limited to a maximum of 8 items borrowed at once.

Librarians manage the system: they add new items to the catalog, register new members, issue loans, accept returns, and calculate fines. The library also tracks reservations, a member can reserve an item that is currently borrowed, and they are notified when it becomes available.

Question: Read the Online Library Management System scenario given above and answer the following questions.

1. Identify all candidate conceptual classes from the scenario above.
2. From your identified candidate conceptual classes, identify any non-classes, subtypes or redundant concepts that you would not retain as a conceptual class and justify each briefly.
3. Propose three associations that are critical for enforcing key business rules in the scenario. And for each:
 - a. Name the two conceptual classes involved.
 - b. Specify the role and multiplicity of the association.
 - c. Explain in one sentence which business rule it helps enforce.
4. Create a complete UML Domain Model Diagram that includes, at least 8 conceptual classes, attributes for 4 of them and five associations with correct multiplicities.
5. The library wants to host events (book clubs, author talks, workshops) for members to attend. Suggest one new conceptual class to add to the model. List its key attributes and describe one association connecting it to an existing class.